Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Assignment - 1

Full Marks: 50 *Deadline: 11:59 PM, 11-08-2025*

Submission Instructions:

1) Answer all the questions.
2) Execute all the commands in Linux and take the screenshots.
3) Prepare a PDF that contains the question number and corresponding screenshots of the outputs.
4) The naming of the file should strictly follow the given format:
   <Roll_No>_<Name>_<Activity_No>
   If your name is Shareen, your roll number is B24CS055, then the filename should be:
   B24CS055_Shareen_02.pdf
5) **Plagiarism is strictly prohibited.**

Objectives:

1) Getting Familiar with Linux Commands.
2) Learning the implementation of Linux commands using C program.
3) Piping and usage of it to manipulate the dataset in different ways.
4) Learning the usage of advanced Linux shell commands.

Task 1: [10 Marks]

[Instruction: If the file is being modified, you have to show the contents of the file with the cat command]

Create a directory named assignment_01_<Your Roll No>. Inside that directory, create a file named weather_details.txt with the following data, where each record should be of the format City Name, Temperature, Humidity, Weather Condition, Wind Speed.

City	Temperature (°C)	Humidity (%)	Weather Condition	Wind Speed (km/h)
New York	28	65	Partly Cloudy	14
Tokyo	31	72	Sunny	10
London	22	80	Rainy	18
Mumbai	30	85	Humid	12
Paris	26	70	Clear	8
Sydney	18	60	Windy	22
Dubai	41	40	Hot	20
Moscow	19	75	Overcast	15
Cape Town	24	55	Sunny	13
Sao Paulo	27	68	Cloudy	11

The data should be appended in the following format:
New York,28,65,Partly Cloudy,14
Tokyo,31,72,Sunny,10
…
Sao Paulo,27,68,Cloudy,11

Now execute the following instructions on the data:

1. Append the city names, temperature and weather conditions whose humidity level is above or equal to 70% and wind speed is greater than or equal to 15 km/h to a file named storm_probability.txt. [2 Marks]
2. Now select the name of the city from the storm_probability.txt who has the lowest temperature and delete the record of the city from the weather_details.txt file and make the modification permanent to the file. [2 Marks]
3. Append all the city names, temperature and wind speed with the format city_name:temperature:wind_speed in the descending order of the temperature to a file cold_weather_high_wind.txt where temperature is less than 25 degrees and wind_speed is "HIGH WIND" if wind speed is greater than or equal to 15 km/h, otherwise "LOW WIND". [3 Marks]
4. Now get the city name from the cold_weather_high_wind.txt whose wind_speed is high and temperature is lowest, and delete the record of the city from the weather_details.txt file and make the modification permanent. [3 Marks]

Task 2: [10 Marks]

Now create a file named file1.txt with the following data:
This is line1.
This is line2.
This is line3.
This is line4.
This is line5.
This is line6.
This is line7.
This is line8.
This is line9.
This is line10.
Now write a C program to make an executable file which takes a number of lines (n) and a filename as arguments and prints those last n lines on the screen. Your filename should be tail.c and the executable file name should be tail.
Sample input: ./tail 3 file1.txt
Sample output:
This is line8.
This is line9.

This is line10.

Task 3: [15 Marks]

1. Create a directory EmployeeRecords in your home directory and navigate into it. [0.5 mark]
2. Inside this directory, create two files, emp_basic.txt and emp_salary.txt. [0.5 mark]
3. Add the following data into emp_basic.txt: [0.5 mark]

   101,John,Manager
   102,Alice,Developer
   103,Bob,Analyst
4. Add the following to emp_salary.txt: [0.5 mark]

   101,85000
   102,70000
   103,65000
5. Modify permissions of both files so that only the owner can view and modify them, while no access is given to others. [1 Mark]
6. From the first file named emp_basic.txt, extract and display only the employee names. [1 Mark]
7. Display emp_salary.txt file contents in descending order based on the salary values and do not modify file content [1 Mark]
8. Merge the data from both files emp_basic.txt and emp_salary.txt based on the employee ID to form a single dataset named merged.txt. [1 Mark]
9. From the merged.txt dataset, extract only the employee names and their salaries. [1 Mark]
10. Replace all instances of the word “Manager” with “Team_Lead” in the first file emp_basic.txt. [1 Mark]
11. Display the total number of lines, words, and characters present in the merged.txt dataset. [1 Mark]
12. Horizontally combine and display the contents of both files emp_basic.txt and emp_salary.txt line by line (i.e., fields appear side by side). [1 Mark]
13. Identify and display all lines from the emp_basic.txt file where the job title ends with the letters “er”. [1 Mark]

14. Create a processing pipeline that:
    - Merges the two datasets emp_basic.txt and emp_salary.txt based on the employee ID
    - Filters out employees earning more than 65,000
    - Sorts the result by salary in descending order
    - Save the above final filtered and sorted output into a new file called high_earners.txt.

    [3 Mark]
15. Identify all employee names that start with a capital letter and have more than three letters. [1 Mark]

Task 4: [15 Marks]

You are working in a university's Linux server to process student academic and stipend details. You have been given two datasets to work with:

File 1 – student_academic.txt (Student ID, Name, Dept, Subject1, Subject2, Subject3)
101,Rahul,CS,78,85,90
102,Aditi,IT,55,60,58
103,Neha,EE,88,92,80
104,Arjun,CS,35,40,45
105,Meera,IT,95,96,94

File 2 – student_stipend.txt (Student ID, Monthly_Stipend)
101,1500
102,0
103,2000
104,0
105,2500

1. Create a directory UniversityDB in your home directory, navigate into it, and create the above two files with the given data. [1 Mark]

2. Set permissions so that: [1 Mark]

   a. Owner: read & write
   b. Group: read only
   c. Others: no access
3. Display the permission details of both files. [1 Mark]
4. Display the current date & time in the format YYYY-MM-DD HH:MM:SS. [1 Mark]
5. Merge both files on Student ID into merged.txt, sorted by ID. [1 Mark]
6. From merged.txt, display only the names and total marks of each student. [1 Mark]
7. Find the topper (student with the highest total marks) and display their full details. [1 Mark]
8. List all students who have failed in at least one subject (marks < 40) along with their department. [1 Mark]
9. Replace the department code CS with Computer_Science in the merged file and save it back. [1 Mark]

10. Display the name(s) of all students who get a stipend greater than 2000. [1 Mark]
11. Using a single command pipeline: [2 Mark]
    a. Merge both files
    b. Filter students with stipend > 0
    c. Sort them by stipend in descending order
    d. Display only Name and Stipend
12. Save the final filtered list from step 11 into stipend_list.txt. [1 Mark]
13. From student_academic.txt, find all names starting with N and having more than 4 letters using regex. [2 Marks]